Chapter 18
Recursion

Java How to Program, 11/e

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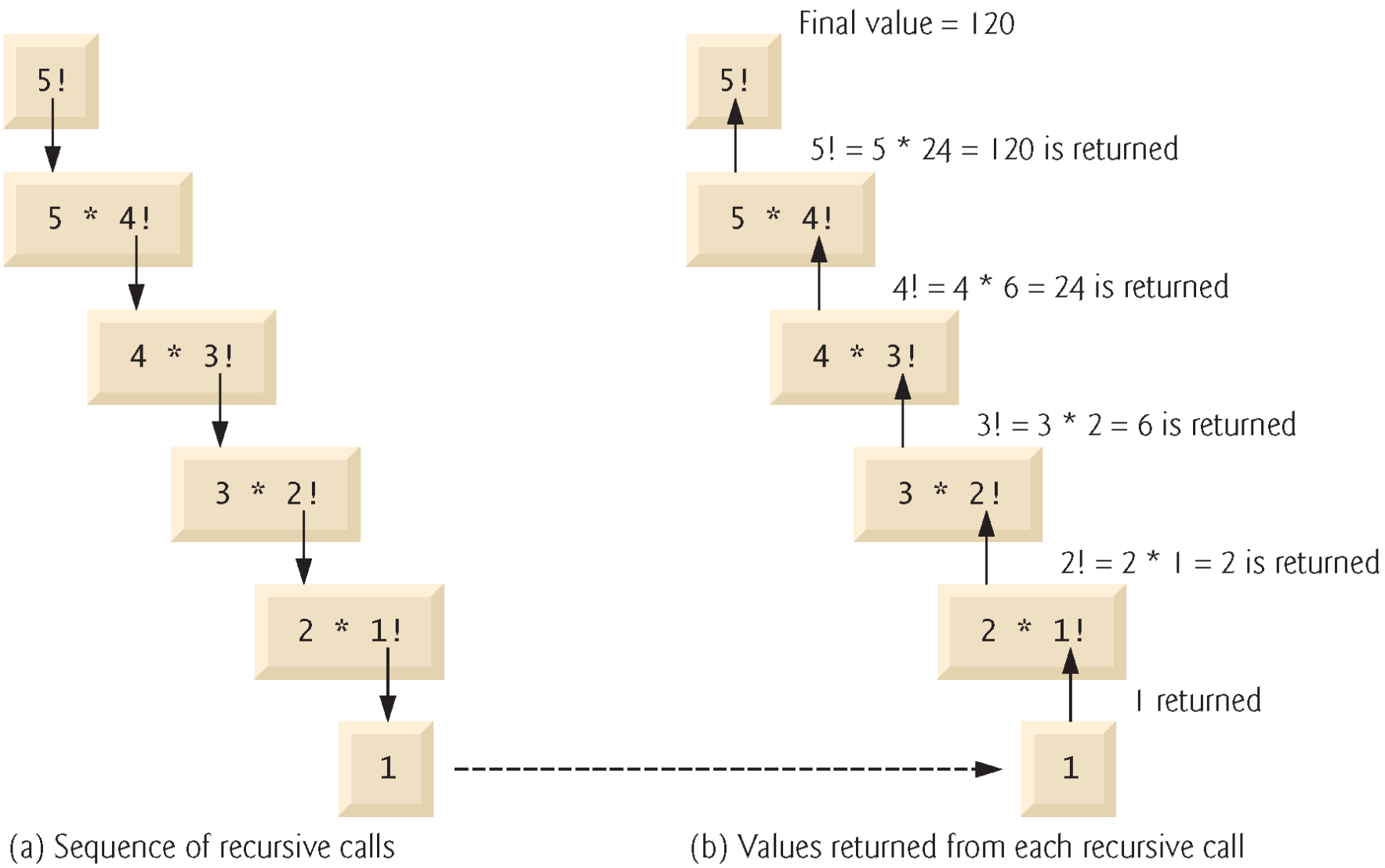

Fig. 18.2 | Recursive evaluation of 5!.

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Common Programming Error 18.1

Either omitting the base case or writing the recursion step incorrectly so that it does not converge on the base case can cause a logic error known as **infinite recursion**, where recursive calls are continuously made until memory is exhausted or the method-call stack overflows. This error is analogous to the problem of an infinite loop in an iterative (nonrecursive) solution.

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```java
// Fig. 18.3: FactorialCalculator.java
// Recursive factorial method.

public class FactorialCalculator {
   // recursive method factorial (assumes its parameter is >= 0)
   public static long factorial(long number) {
      if (number <= 1) { // test for base case
         return 1; // base cases: 0! = 1 and 1! = 1
      }
      else { // recursion step
         return number * factorial(number - 1);
      }
   }
```

Fig. 18.3 | Recursive factorial method. (Part 1 of 2.)

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```java

    public static void main(String[] args) {
        // calculate the factorials of 0 through 21
        for (int counter = 0; counter <= 21; counter++) {
            System.out.printf("%d! = %d%n", counter, factorial(counter));
        }
    }
}
```

```
0! = 1
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
...
12! = 479001600 —— 12! causes overflow for int variables
...
20! = 2432902008176640000
21! = -4249290049419214848 —— 21! causes overflow for long variables
```

Fig. 18.3 | Recursive factorial method. (Part 2 of 2.)

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```java
// Fig. 18.4: FactorialCalculator.java
// Recursive factorial method.
import java.math.BigInteger;

public class FactorialCalculator {
   // recursive method factorial (assumes its parameter is >= 0)
   public static BigInteger factorial(BigInteger number) {
      if (number.compareTo(BigInteger.ONE) <= 0) { // test base case
         return BigInteger.ONE; // base cases: 0! = 1 and 1! = 1
      }
      else { // recursion step
         return number.multiply(
            factorial(number.subtract(BigInteger.ONE)));
      }
   }
```

Fig. 18.4 | Factorial calculations with a recursive method. (Part 1 of 2.)

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```java

    public static void main(String[] args) {
       // calculate the factorials of 0 through 50
       for (int counter = 0; counter <= 50; counter++) {
          System.out.printf("%d! = %d%n", counter,
             factorial(BigInteger.valueOf(counter)));
       }
    }
 }
```

```
0! = 1
1! = 1
2! = 2
3! = 6
...
21! = 51090942171709440000 —— 21! and larger values no longer cause overflow
22! = 1124000727777607680000
...
47! = 258623241511168180642964355153611979969197632389120000000000
48! = 12413915592536072670862289047373375038521486354677760000000000
49! = 608281864034267560872252163321295376887552831379210240000000000
50! = 30414093201713378043612608166064768844377641568960512000000000000
```

Fig. 18.4 | Factorial calculations with a recursive method. (Part 2 of 2.)

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```java
// Fig. 18.5: FibonacciCalculator.java
// Recursive fibonacci method.
import java.math.BigInteger;

public class FibonacciCalculator {
   private static BigInteger TWO = BigInteger.valueOf(2);

   // recursive declaration of method fibonacci
   public static BigInteger fibonacci(BigInteger number) {
      if (number.equals(BigInteger.ZERO) ||
            number.equals(BigInteger.ONE)) { // base cases
         return number;
      }
      else { // recursion step
         return fibonacci(number.subtract(BigInteger.ONE)).add(
            fibonacci(number.subtract(TWO)));
      }
   }
```

Fig. 18.5 | Recursive fibonacci method. (Part 1 of 3.)

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```java

    public static void main(String[] args) {
        // displays the fibonacci values from 0-40
        for (int counter = 0; counter <= 40; counter++) {
            System.out.printf("Fibonacci of %d is: %d%n", counter,
              fibonacci(BigInteger.valueOf(counter)));
        }
    }
}
```

Fig. 18.5 | Recursive fibonacci method. (Part 2 of 3.)

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```
Fibonacci of 0 is: 0
Fibonacci of 1 is: 1
Fibonacci of 2 is: 1
Fibonacci of 3 is: 2
Fibonacci of 4 is: 3
Fibonacci of 5 is: 5
Fibonacci of 6 is: 8
Fibonacci of 7 is: 13
Fibonacci of 8 is: 21
Fibonacci of 9 is: 34
Fibonacci of 10 is: 55
...
Fibonacci of 37 is: 24157817
Fibonacci of 38 is: 39088169
Fibonacci of 39 is: 63245986
Fibonacci of 40 is: 102334155
```

Fig. 18.5 | Recursive fibonacci method. (Part 3 of 3.)

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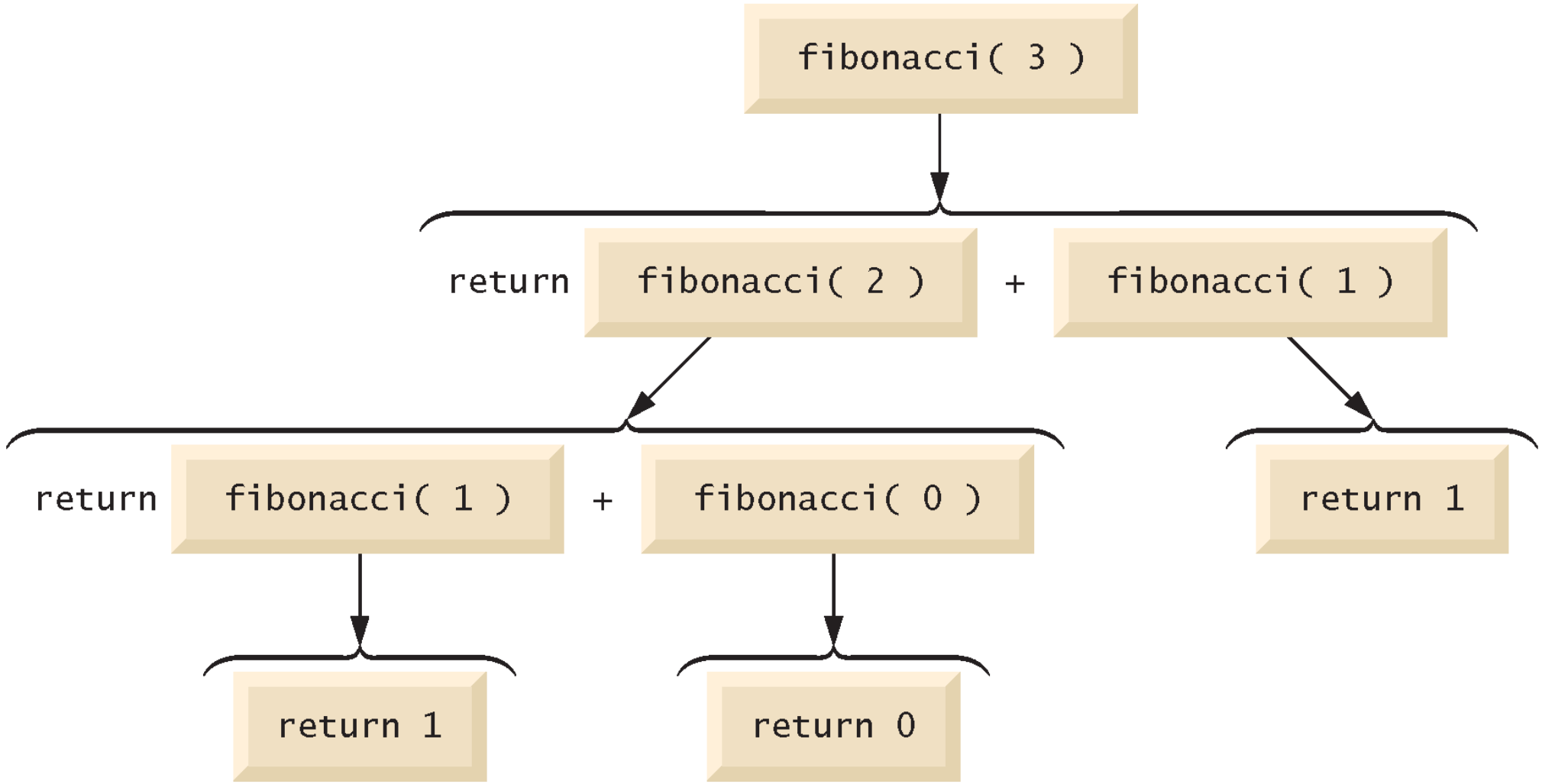

Fig. 18.6 | Set of recursive calls for `fibonacci(3)`.

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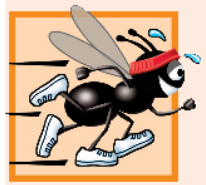

Performance Tip 18.1

Avoid Fibonacci-style recursive programs, because they result in an exponential "explosion" of method calls.

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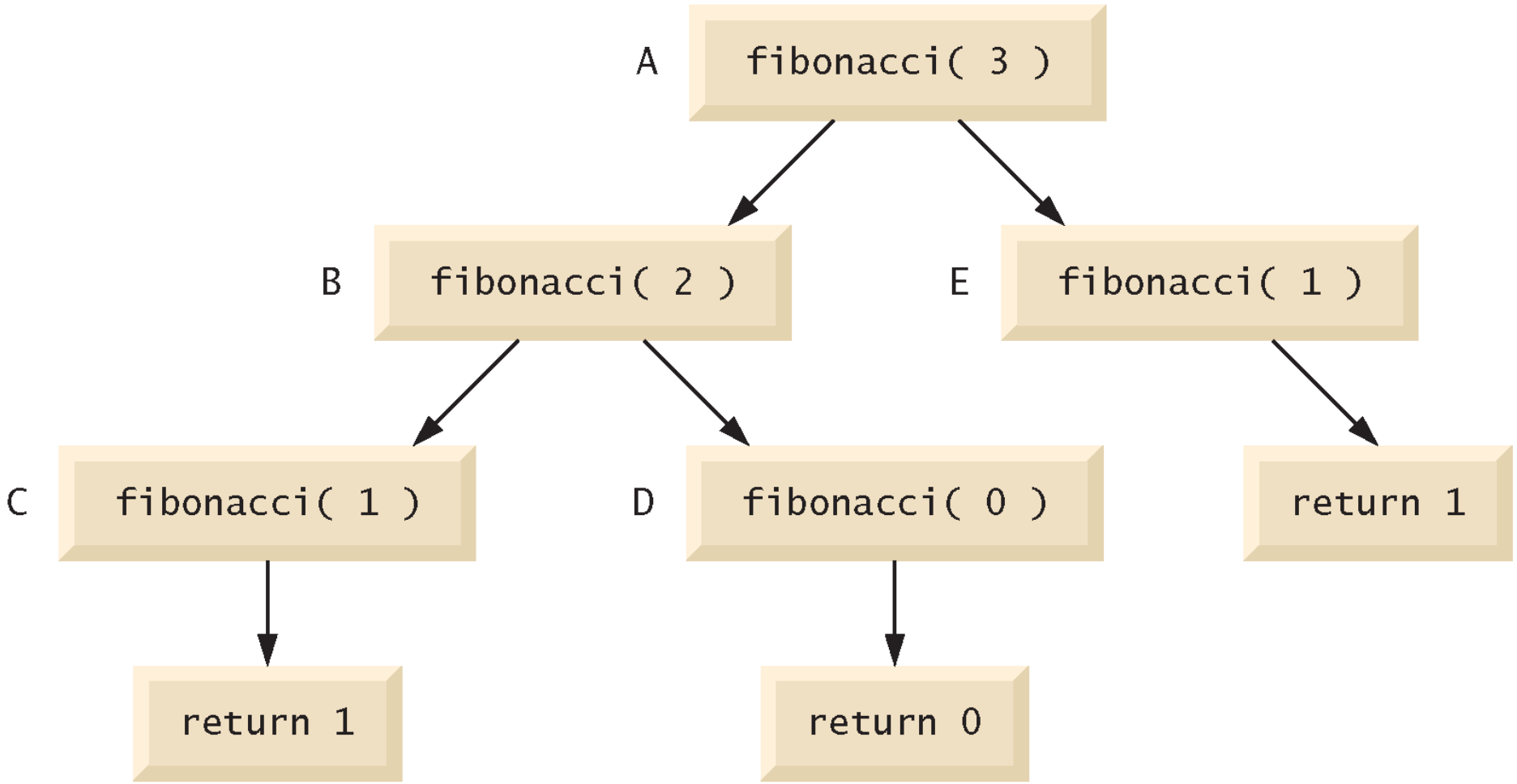

Fig. 18.7 | Method calls made within the call `fibonacci(3)`.

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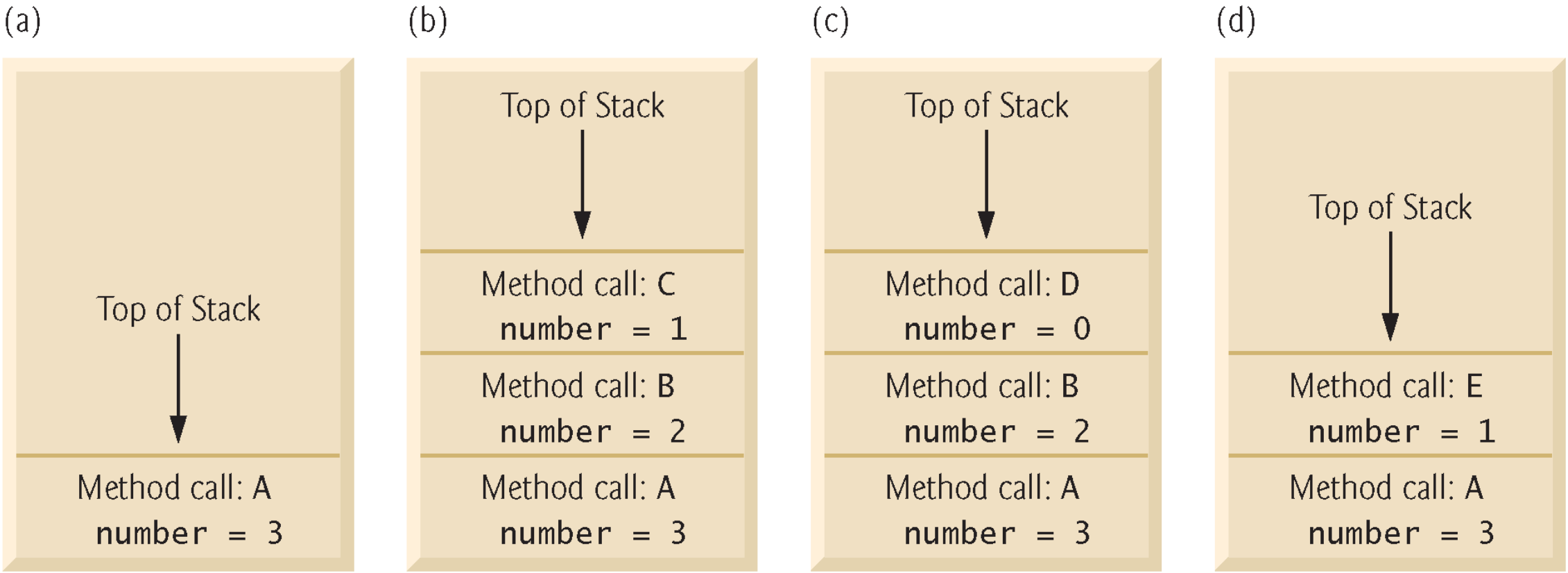

Fig. 18.8 | Method calls on the program-execution stack.

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```java
// Fig. 18.9: FactorialCalculator.java
// Iterative factorial method.

public class FactorialCalculator {
   // iterative declaration of method factorial
   public long factorial(long number) {
      long result = 1;

      // iteratively calculate factorial
      for (long i = number; i >= 1; i--) {
         result *= i;
      }

      return result;
   }

```

Fig. 18.9 | Iterative factorial method. (Part 1 of 2.)

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```java
    public static void main(String[] args) {
        // calculate the factorials of 0 through 10
        for (int counter = 0; counter <= 10; counter++) {
            System.out.printf("%d! = %d%n", counter, factorial(counter));
        }
    }
}
```

```
0! = 1
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
6! = 720
7! = 5040
8! = 40320
9! = 362880
10! = 3628800
```

Fig. 18.9 | Iterative factorial method. (Part 2 of 2.)

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Software Engineering Observation 18.1

Any problem that can be solved recursively can be solved iteratively and vice versa. A recursive approach is preferred over an iterative approach when the recursive approach more naturally mirrors the problem and results in a program that's easier to understand and debug. A recursive approach can often be implemented with fewer lines of code. Another reason to choose a recursive approach is that an iterative one might not be apparent.

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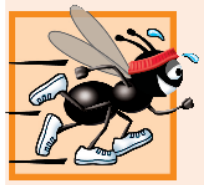

Performance Tip 18.2

When performance is crucial, you might want to try various iterative and recursive approaches to see which achieve your goal.

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Common Programming Error 18.2

Accidentally having a nonrecursive method call itself either directly or indirectly through another method can cause infinite recursion.

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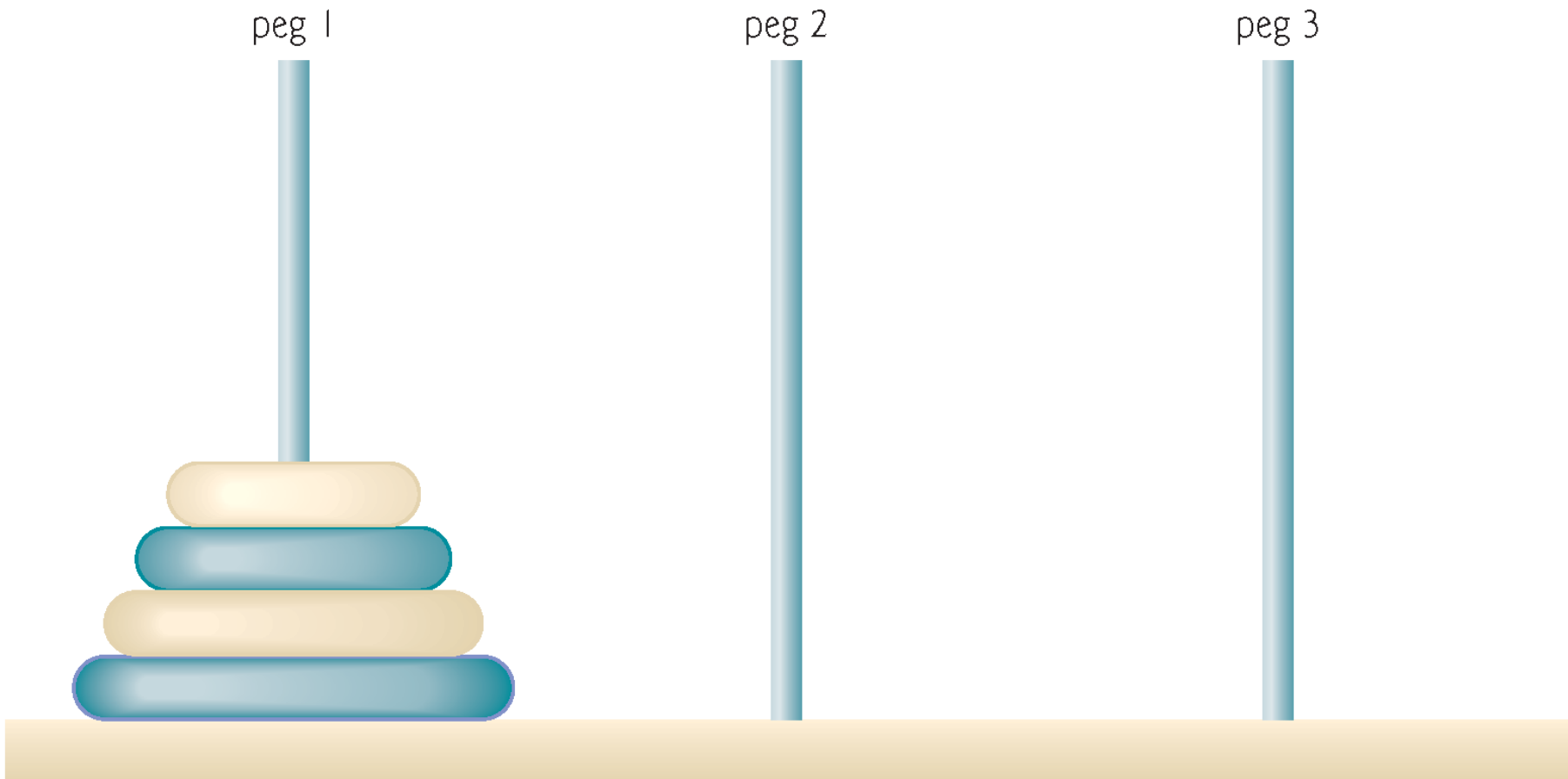

Fig. 18.10 | Towers of Hanoi for the case with four disks.

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```java
// Fig. 18.11: TowersOfHanoi.java
// Towers of Hanoi solution with a recursive method.
public class TowersOfHanoi {
   // recursively move disks between towers
   public static void solveTowers(int disks, int sourcePeg,
      int destinationPeg, int tempPeg) {
      // base case -- only one disk to move
      if (disks == 1) {
         System.out.printf("%n%d --> %d", sourcePeg, destinationPeg);
         return;
      }

      // recursion step -- move (disk - 1) disks from sourcePeg
      // to tempPeg using destinationPeg
      solveTowers(disks - 1, sourcePeg, tempPeg, destinationPeg);

      // move last disk from sourcePeg to destinationPeg
      System.out.printf("%n%d --> %d", sourcePeg, destinationPeg);

      // move (disks - 1) disks from tempPeg to destinationPeg
      solveTowers(disks - 1, tempPeg, destinationPeg, sourcePeg);
   }
```

Fig. 18.11 | Towers of Hanoi solution with a recursive method. (Part 1 of 2.)

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